

Safal Shrestha

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Education

Bachelor of Science in Computer Science | Expected May 2026

University of Idaho | Moscow, ID

Relevant Coursework:

- Computational Biology: Sequence Alignment
- Discrete Mathematics & Calculus
- Analysis of Algorithms
- Linear Algebra

Skills

- Programming Languages: C/C++, Python, Java, SQL
- Web Development: HTML, CSS, JavaScript, React, Node
- Tools & Software: Git/GitHub, Microsoft Office, Power BI, Tableau, Matplotlib, NumPy
- IT & Networking: Basic server maintenance, vulnerability scanning, Debugging

Experience

Programmer | anka Ek | 02/2018 - 11/2021

- Built a full-stack e-commerce platform using HTML, CSS, and JavaScript, ensuring a responsive and visually appealing user experience
- Implemented authentication and payment integration by working with backend developers to create seamless user interactions.
- Collaborated with UI/UX designers to implement interactive web pages, improving accessibility and usability.
- Redesigned the website to create a more visually appealing, user-friendly experience.

Supervisor of Operations | Compass Group | University of Idaho | Moscow, ID | 02/2022 – 01/2025

- Managed the work of 45 employees, increasing crew efficiency by 15%.
- Developed a presentation and training manual for staff to use in customer service interaction.
- Trained 15-20 new employees in customer service script recitation, conflict resolution, and data entry practices.

Project

Genetic Algorithm | University of Idaho

- Designed and implemented a genetic algorithm (GA) to analyze the effect of mutation rate on evolutionary performance.
- Conducted experiments by varying mutation rates across multiple runs, measuring impact on fitness improvement speed, maximum fitness achieved, and population diversity.
- Visualized data using Matplotlib and performed statistical analysis with NumPy.
- Implemented the project in Python and structured the code for clarity and extensibility.

Programming Language Development Project | University of Idaho

- Created a custom programming language from scratch using Java, with a simple C-like syntax.
- Built a lexer, parser, and interpreter to process and execute code.
- Added basic features like variables, loops, conditionals, and arithmetic operations.
- Designed an error-handling system to catch and explain coding mistakes.
- Optimized performance to ensure smooth execution.
- Wrote documentation and test cases to ensure reliability.

Movie Recommendation System | University of Idaho

- Built a Movie Recommendation System that suggests movies based on user preferences, ratings, and watch history.
- Developed a recommendation engine using collaborative and content-based filtering to improve movie suggestions.
- Created a user-friendly web interface where users can create profiles, rate movies, and manage their watchlists.
- Integrated IMDb and TMDb APIs to fetch real-time movie details like ratings and genres.
- Used Java with Spring Boot for the backend and React.js for the frontend.

